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EB Programmer

Version 1.7.2

User guide

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Contents

1 Introduction

1.1 Features

EB Programmer is a application for PC dedicated for communication with EB-3 Demo Board. It allow to transmit the user executable program as memory content to the EB-3 and to monitor running the software on EB-3 om EB-3 Demo Board.

EB Programmer also provide tools to check that EB-3 equipment is working properly.

1.2 Target audience

This manual is intended for designers who use, or consider to use the EB Programmer application with the EB-3 Demo Board.

1.3 Installing the EB Programmer

Run the **setup.exe** from install directory and follow program installer instructions.

1.4 Modifications history

Version 1.6	- December 2000
Version 1.7.0	- September 2001
Version 1.7.1	- November 2001

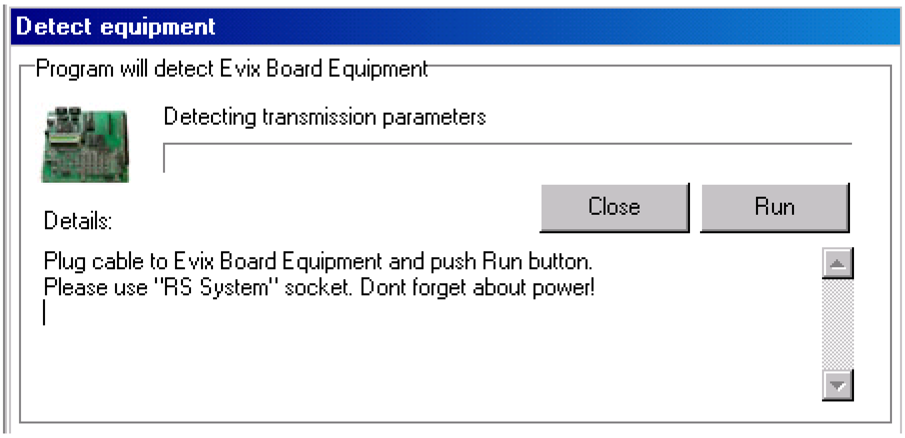
In respect version 1.6 the Dhrystone and Whetstone benchmark tests are added and the Automatic detection feature.

In respect version 1.7.0 the Whetstone benchmark tests is added.

2 User guide

2.1 Start

When the EB Programmer is launched program will run the automatic detection of equipment.



Press the **Run** button for start the automatic detection or **Close** button for manually setup.

2.2 Automatic detection

Automatic detection is a function that automatically check the EB-3 Demo Board is properly connected to ports and configure a setup for transmission.

During auto detection, program will check all existing COM ports of PC. Program will check free ports by sending request to equipment. If equipment connected to the port will correct answer then automatic detection process stops and display the message:

Equipment ready to work!

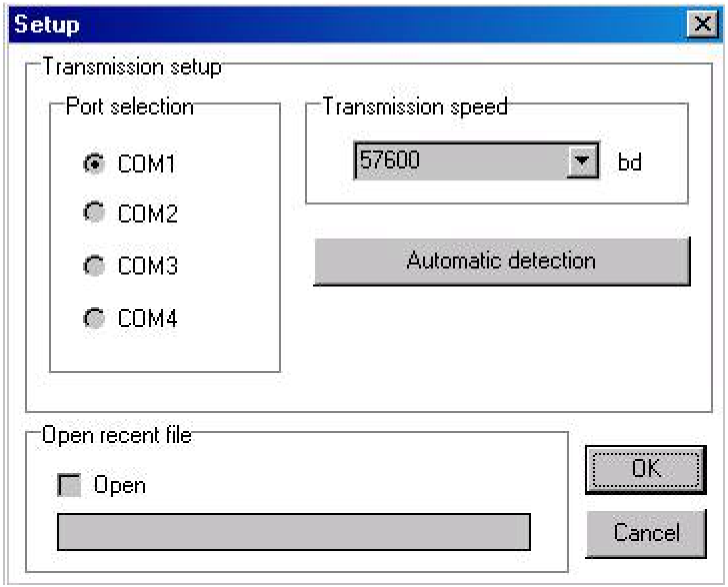
Correct configuration is saved in system registry and after next start, software will be try connect with equipment using saved configuration. If this check fail then program will show again a automatic detection window.

To run the automatic detection function open the setup window from **File/Setup** menu and press the **Automatic Detection** button.

2.3 Setup

a) Setup window

Choose a **File** menu and then a **Setup** bar to open a **Setup** window.



b) Transmission setup

Choose one of the existing COM ports in a **Ports selection.** If a selected port is busy or does not exist, system will warn you:

Error during access to port!

You can also run an **Automatic detection** option for an automatic configuration. The **Automatic detection** require connecting to the EB-3 Demo Board.

c) Open recent file

If an **Open** box is selected then system opens the last recent file at startup. This file is specified in a 'Recent file' box.

d) Transmission speed

Allows to set a transmission speed range from 110 bauds to 256000 bauds. The recommended speed is 9600 bauds or 57600 bauds, depending the EB-3 Demo Board version.

2.4 Run user application

To load your application and send it to the EB-3 follow the steps below.

e) Read the file

Next choose a **Read file** in the **File** menu to load your program file into viewer. Select and open the file using the **Open** window. The EB Programmer reads two memory content file formats: INTEL hex format or Motorola S Record. The limitation for read format are:

- Intel hex format – only 8 bit Data and End of File records are acceptable.
- Motorola S Record – only 8 bit S1 records are acceptable.

f) Define a target type

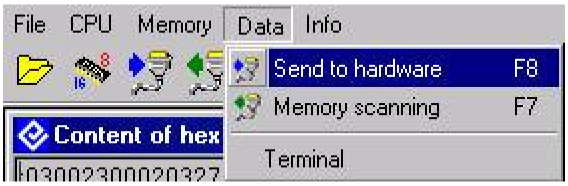
EB-3 Demo Board supports 8-bit or 16-bit processors. That is the reason why user should set kind of controller to give to equipment a information how store data in memory during transmission.

To define target type select right family by choosing **Type** in the **CPU** menu. Please refer to chapter 3.2 for more information.

You can select **Free setup** type and memory width (8-bit or 16-bit).

g) Send to EB-3

Since you have opened the file you can send it to the EB-3. Choose a **Send to hardware** in the **Data** main menu to start the transmission.



A properly opened file will inform you:

Transmission completed successfully!

3 Application notes

3.1 How does the transmission work?

During opening a file the EB Programmer reads lines as strings and a line size does not matter. When a S-Record is read, it is converted to an Intel Hex format, because an onboard control processor works only with the Intel Hex.

When an operation **Send to hardware** is chosen then the EB Programmer sends the character with a description of a processor type:

- value **9** to set the **processor support** mode to 8 bit
- value **10** to set the **processor support** mode to 16 bit.

The EB Programmer sends then a single character ':' (58 dec) and waits for an answer from the EB-3 Demo Board. A correct answer from the EB-3 is '>' (62 dec).

The EB Programmer sends then a single line and waits again for an answer form board, the correct answer is '>' (62 dec). The EB-3 control processor sends the answer ('>' 62) to the EB Programmer after checking the CRC. If the CRC is correct the EB Programmer sends the next ':' character and waits for an answer ('>'). Then sends another line, etc.

Afterwards, you can be sure that the data was correctly stored in the demo board system. If there were no transmission errors but system still doesn't work, the problem may be caused by a user program or a hardware board error (data, address bus).

3.2 Target - Processor type

The EB-3 Demo Board can work with some of 8-bit or 16-bit processors. It depends on the EB-3 firmware (BIOS, LOADER, GAL) adequately to the kind of processor working in EB-3 Demo Board.

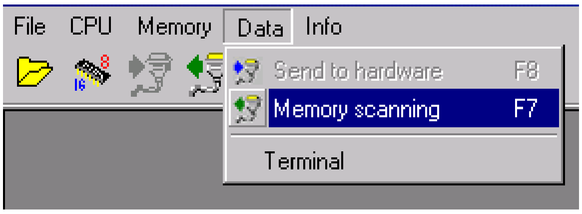
The format of the data transmission between PC and the EB-3 is always Intel Hex. That is why the EB Programmer converts the other data formats (e.g. Motorola S Record) to Intel Hex before transmitting them to the EB-3. To use the EB-3 with a 16-bit processor, set your compiler to an 8-bit output mode.

For a proper operation the EB-3 needs information about a processor type to correctly store data in memory. Choose processor type in dialog box. Each listed processors have assigned width 8-bit or 16-bit of the data and address busses. It is available to chose a **Free setup** to manually set target type (8-bit or 16-bit) and order of bytes. Sometimes it is necessary to swap bytes in a 16-bit word. In order to do this choose an **Invert bytes order**.

4 Evaluation tools

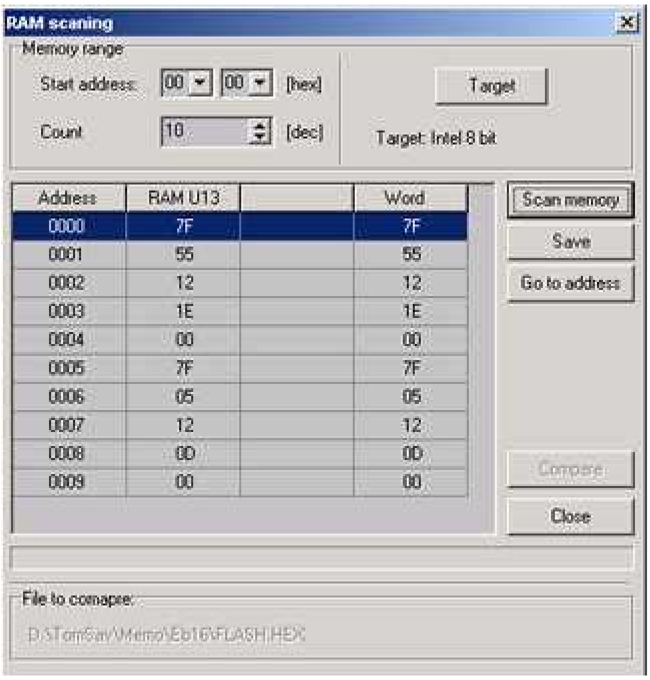
4.1 Memory scanning

It is possible to use the EB-3 Demo Board and the EB Programmer to download RAM contents. The procedure reads a block of RAM beginning at a **Start address** until the end of a block. The length of memory block depends on a **Count** parameter. To run a memory scanner choose a **Memory scanning** in the **Data** main menu or a keyboard shortcut F7.



Set a **Start address** to determine a begin address. Set a **Count** to determine a number of words to be downloaded. Set an appropriate **Target** processor type to **8 bit** or **16 bit**.

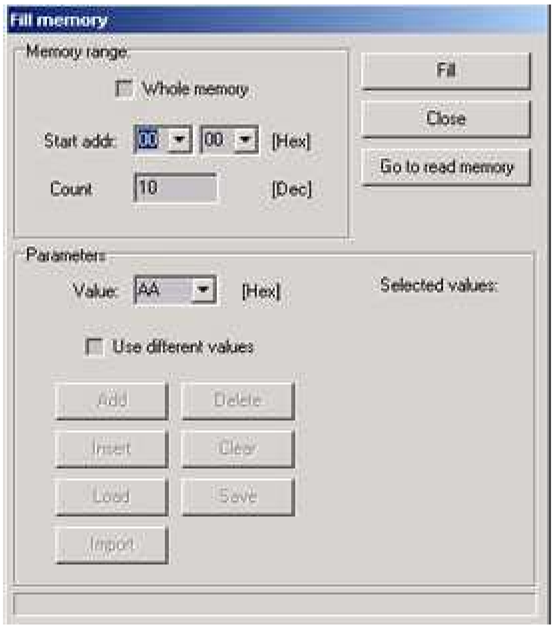
Press a **Scan memory** to run memory scanning.



After downloading the data you are able to look through the memory contents or save them into the file by choosing a **Save** button.

4.2 Fill memory

You can fill a block of memory with a specified value without use of a hex file. Use a **Fill memory** in the **Memory** main menu.



In a **Memory range** section you can select a **Whole memory**, or specify a range of **Count** words beginning at a **Start address**.

Then specify a value to fill in. If an **Use different values** is checked, it is possible to determine a list of values to be sent to memory. You can apply values by using an **Add** button, or delete any value by clicking on it (on the list) and choosing a **Delete**.

It is also possible to save or load a list of values with use of **Save** and **Load** buttons or import a list stored in the text file (**Import** button).

This are the sample data prepared for import (values should be like those below):

```
7F03 5EFA CDAB21436587BF3F5EFACDAB00
2143 6587 C03F5EFACDAB21436587C13F00
5EFA CDAB 21436587C23F5EFACDAB214300
6587 C33F 5EFACDAB21436587C43F5EFA00
```

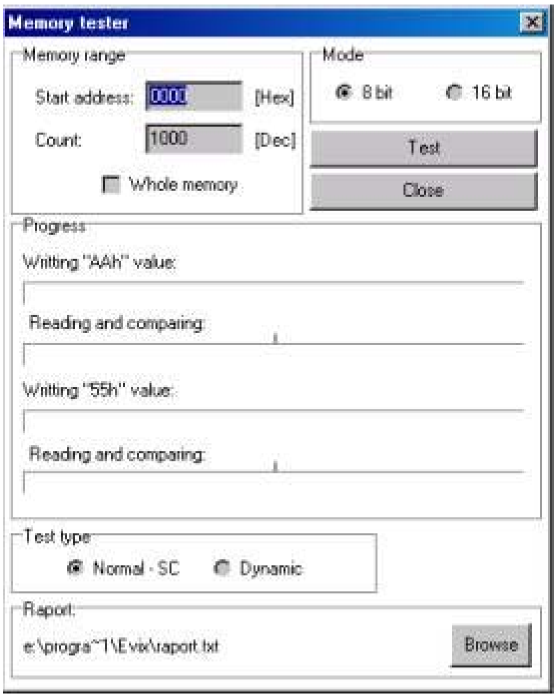
Spaces between characters are allowed.

4.3 Test memory

Before start working with the EB-3, use a **Test memory** procedure. This allows you to check the Data and Address buses located on board. If there are any chips or tracks defects, it will be reported.

This tool and the CRC protected transmission makes you sure that your hex file is correctly transmitted to memory.

To run a board test, select a **Memory Test** in the **Memory** main menu. Then a **Memory tester** dialog window will occur.



Now you can set a memory range to check the **Whole memory** or a specific range form **Start address** by **Count** memory words. Choose a memory test type: **Normal** or **Dynamic**. During a **Normal** test to the memory will be written $AA (hex) value, then read and compared with $AA. The next memory test will be written with $55 (hex) value, then read and compared with $55.

During a **Dynamic** test to the memory will be written sequence off values: 24h, 49h, 92h, 24h, 49h, 92h …next memory is read and compared with a defined sequence.

To run the test push a **Test** button. Test results will be written into a text file.

In case of problems with communication with EB-3 Demo Board or memory test fails then remove FPGA adapter with IP Core and try run tests again. If it will succeed then there are two possibilities:

- The FPGA adapter is damaged or
- The microprocessor core is not halted in reset state during transmission

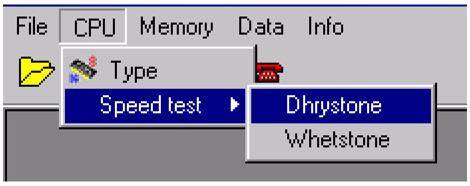
4.4 Speed test

The speed test allow to run the Dhrystone or Whetstone test with differ versions of microprocessor core.

To run speed tests at first choose the processor type in main menu **CPU -> Type** and then select right CPU type as for example: Intel 8051 or Motorola 68k. Do not chose **Free setup**, because with each kind of processor have assigned special benchmark code, and software have to know which file should be loaded for transmission during an speed test.

After that hit in main menu **CPU -> Speed** test and chose a kind of test:

- Dhrystone for integer operations
- Whetstone for floating point operations



EB Programmer will transmit a test program to EB-3. Tests program are installed during EB Programmer installation in subfolder **\CPUTest**.

Contents of folder depends of processor version. It could be possible that in your current version is not included one or more of listed files.

For Intel 8051 compatible:

- 8051dhry.hex
- 8051whet.hex

For Zilog Z80 compatible:

- Z80dhry.hex
- Z80whet.hex

For Motorola 68000 compatible:

- M68Kdhry.hex
- M68Kwhet.hex

For Intel 186 compatible:

- 80186dhry.hex
- 80186whet.hex

For TMS 320 compatible:

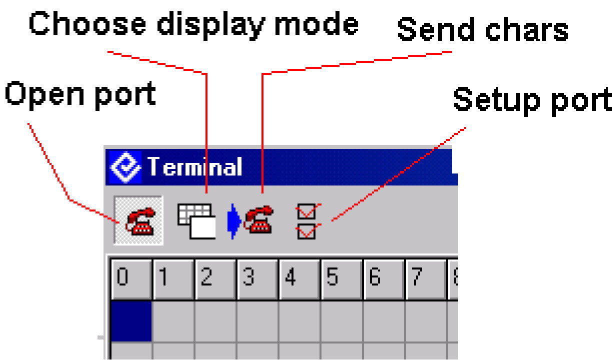
- TMS320dhry.hex
- TMS320whet.hex

After transmission of test program, application will open the **Terminal** window. In this window will be displayed (it can take w while) an text as example:

```
************************************
Now running 5000 Dhrystones...
Dhrystone(1.1) time for 5000 passes = 3141
This machine benchmarks at 198 dhrystones/second
Bench time=3141   Null time=31
************************************
```


4.5 Serial port terminal

The Terminal is a tool that can read and write PC serial ports in all speed modes. To run Terminal hit in main menu **Data –> Terminal**. The Terminal is displaying a transmission data in two different modes: the test mode or characters mode.



To set up port used by Terminal chose a **Setup port** button on tool bar, then on special dialog box please chose a port and transmission speed. The transmission speed of Terminal is independent the main setup. You can use with Terminal different COM port or different then main setup. If you select **Main** setup then Terminal will use the same settings as main EB Programmer setup.

Terminal can send a data trough the port, to send data please write characters on edit box on bottom of Terminal window and next hit **Send** chars button on tool bar.

5 Support

5.1 Information

The basic info is available in a **Help** bar of the **Info** menu.

Please visit also our web site www.evatronix.pl for Evatronix information.

There is also service with information available on-line about history of modifications and for downloading the updates for the EB Programmer. Please visit the web site of the Software Support Center by choosing in main menu **Info -> Look for update**.

5.2 Technical support

In case of any problems please contact:

Evatronix Technical support:
ipcenter@evatronix.com.pl
or

EB-3 Design Engineer:
Wojciech Pietrasina: wojpie@bielsko.evatronix.com.pl
EB Programmer Software Engineer:
Tomasz Jakobiec - tomjak@bielsko.evatronix.com.pl